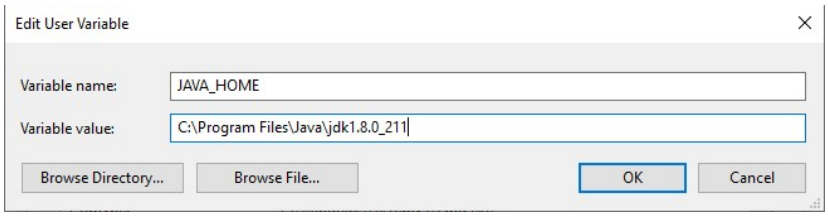
Setup Python Virtual Environment to run PySpark in VSCode

Install Prerequisites

1. **Make sure you are running Java 8 (JDK 1.8.xxx) or up. Install if required.**
   - java -version
   - URL: https://www.oracle.com/in/java/technologies/downloads/#jdk24-windows

2. **Add JAVA_HOME as an environment variable**



3. **Make sure a suitable version of python is installed on your machine**
   a. Check version compatibility using the following link
   b. https://community.cloudera.com/t5/Community-Articles/Spark-Python-Supportability-Matrix/ta-p/379144

Setup Python Virtual Environment (venv)

4. **Open command prompt (or PowerShell) on your windows machine and run the following commands to create a python virtual environment in your project folder.**

   a. Create a project folder in which you want to install the venv (virtual environment)
      - `mkdir pyspark-project`

   b. Create a python virtual environment (folder: **pyspark-project**, venv name: **venv**)
      - ```
        cd pyspark-project
        python -m venv venv
        ```

   c. Activate the virtual environment.
      - `venv\Scripts\Activate.ps1  => Power Shell`
      - `venv\Scripts\activate  => Command prompt`

5. **Install pyspark in the virtual environment and launch VSCode**

   a. Install PySpark using pip in your virtual environment
        ```

- **`pip install pyspark`**

b. Launch VSCode
- **`code .`**

```
E:\PySpark\vscode>python -m venv venv

E:\PySpark\vscode>venv\Scripts\activate.bat

(venv) E:\PySpark\vscode>pip install pyspark

(venv) E:\PySpark\vscode>code .
```

6. **Open your PySpark scripts in VSCode and execute the program.**
   Note: If you are running the scripts provided by me, make sure you delete the first two lines in the code related to importing and initializing `findspark` library.
   a. You can run the python script in the VSCode code terminal using the following command:
      i. `python <script-path>`
   b. If you want to run specific lines of python code, you can select the code and use Shift+Enter keyboard shortcut.

Common Issues & Fixes

Issue	Fix
`JAVA_HOME not set` error	Recheck Step 2, restart VS Code fully
`winutils.exe` / Hadoop errors on Windows	Download `winutils.exe` matching your Hadoop version, place in `C:\hadoop\bin`, set `HADOOP_HOME=C:\hadoop`
Pylance can't resolve `pyspark` import	Confirm correct interpreter selected (Step 7); reload window (`Ctrl+Shift+P` → Reload Window)
PowerShell activation blocked	Run the `Set-ExecutionPolicy` command from Step 3